

# Project1 solution

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## The code:

```
print("Welcome to Tic Tac Toe Game")
import random

def display(board):
    print(board[0], "|", board[1], "|", board[2])
    print("-----")
    print(board[3], "|", board[4], "|", board[5])
    print("-----")
    print(board[6], "|", board[7], "|", board[8])

#check win
def win(board, player):
    return (
        (board[0]==board[1]==board[2]==player) or
        (board[3]==board[4]==board[5]==player) or
        (board[6]==board[7]==board[8]==player) or
        (board[0]==board[3]==board[6]==player) or
        (board[1]==board[4]==board[7]==player) or
        (board[2]==board[5]==board[8]==player) or
        (board[0]==board[4]==board[8]==player) or
        (board[2]==board[4]==board[6]==player)
    )

#check draw
def draw(board):
    return " " not in board

#check free pos
def free(board, pos):
    return board[pos] == " "

def computer_move(board):
    available = [i for i in range(9) if board[i] == " "]
    return random.choice(available)

#Game start here :
while True:    #true until win or draw

    board = [" "] * 9    #new clear board

    mode = input("1- Player  2- Computer: ")
    player1 = input("Choose X or O: ").upper()

    if player1 == "X":
        player2 = "O"
```

```

else:
    player2 = "X"

start = input("Do you want to start? (yes/no): ") #who play first

if start.upper() == "YES":
    current = player1
else:
    current = player2

#
print("Positions are from 0 to 8 :")
print("0 | 1 | 2")
print("3 | 4 | 5")
print("6 | 7 | 8")

while True: #start playing
    display(board)

    #other player take action or computer
    if mode == "1" or current == player1:
        pos = int(input("Choose position (0-8): "))
    else:
        pos = computer_move(board)
        print("Computer chose:", pos)

    #if free pos play, else "not available" & ask for enter another pos
    if free(board, pos):
        board[pos] = current
    else:
        print("Not available")
        continue

    #check win
    if win(board, current):
        display(board)
        print("Winner is", current)
        break

    #check draw
    if draw(board):
        display(board)
        print("Draw!")
        break

    #next turn
    if current == player1:
        current = player2
    else:
        current = player1

again = input("Play again? (yes/no): ")
if again.upper() != "YES":
    break

```

## The input & output:

Welcome to Tic Tac Toe Game

1- Player 2- Computer: 1

Choose X or O: x

Do you want to start? (yes/no): yes

Positions are from 0 to 8 :

```
0 | 1 | 2
3 | 4 | 5
6 | 7 | 8
|   |
```

```
-----
|   |
```

```
-----
|   |
```

Choose position (0-8): 0

```
x |   |
```

```
-----
|   |
```

```
-----
|   |
```

Choose position (0-8): 4

```
x |   |
```

```
-----
| 0 |
```

```
-----
|   |
```

Choose position (0-8): 2

```
x |   | x
```

```
-----
| 0 |
```

```
-----
|   |
```

Choose position (0-8): 1

```
x | 0 | x
```

```
-----
| 0 |
```

```
-----
|   |
```

Choose position (0-8): 7

```
x | 0 | x
```

```
-----
| 0 |
```

```
-----
| x |
```

Choose position (0-8): 3

```
x | 0 | x
```

```
-----
0 | 0 |
```

```
-----
| x |
```

Choose position (0-8): 5

```
x | 0 | x
```

```
-----
0 | 0 | x
```

```
-----
| x |
```

Choose position (0-8): 8

```
X | O | X
```

```
-----
```

```
O | O | X
```

```
-----
```

```
  | X | O
```

Choose position (0-8): 6

```
X | O | X
```

```
-----
```

```
O | O | X
```

```
-----
```

```
X | X | O
```

Draw!

Play again? (yes/no): no

Process finished with exit code 0

---

## More experiences: -

### The input & output:

Welcome to Tic Tac Toe Game

Player 2- Computer: 2 -1

Choose X or O: 0

Do you want to start? (yes/no): yes

Positions are from 0 to 8 :

```
2 | 1 | 0
```

```
5 | 4 | 3
```

```
8 | 7 | 6
```

```
|  |
```

```
-----
```

```
|  |
```

```
-----
```

```
|  |
```

Choose position (0-8): 8

```
|  |
```

```
-----
```

```
|  |
```

```
-----
```

```
0 |  |
```

Computer chose: 5

| |  
-----

X | |  
-----

O | |

Choose position (0-8): 7

| |  
-----

X | |  
-----

O | O |

Computer chose: 6

| |  
-----

X | |  
-----

X | O | O

Choose position (0-8): 3

| |  
-----

O | | X  
-----

X | O | O

Computer chose: 1

X | |  
-----

O | | X  
-----

X | O | O

Choose position (0-8): 1

Not available

X | |  
-----

O | | X  
-----

X | 0 | 0

Choose position (0-8): 2

X | 0 |

-----

0 | | X

-----

X | 0 | 0

Computer chose: 0

X | X | 0

-----

0 | | X

-----

X | 0 | 0

Choose position (0-8): 4

X | X | 0

-----

0 | 0 | X

-----

X | 0 | 0

Draw!

Play again? (yes/no): yes

Player 2- Computer: 2 -1

Choose X or O: X

Do you want to start? (yes/no): yes

Positions are from 0 to 8 :

2 | 1 | 0

5 | 4 | 3

8 | 7 | 6

| |

-----

| |

-----

| |

Choose position (0-8): 8

| |

-----  
| |

-----  
X | |

Computer chose: 3

| |  
-----

O | |  
-----

X | |  
Choose position (0-8): 4

| |  
-----

O | X |  
-----

X | |  
Computer chose: 0

O | |  
-----

O | X |  
-----

X | |  
Choose position (0-8): 6

O | |  
-----

O | X |  
-----

X | | X  
Computer chose: 1

O | O |  
-----

O | X |  
-----

X | | X  
Choose position (0-8): 7

0 | 0 |

-----

0 | x |

-----

x | x | x

Winner is x

Play again? (yes/no): no

Process finished with exit code 0